SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – I : RAILWAY ENGINEERING

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Introduction – Permanent Way Components – Cross Section of Permanent Way – Functions and Requirements of Rails, Sleepers and Ballast – Types of Gauges – Creep of Rails – Theories Related to Creep – Coning of Wheels – Adzing of Sleepers – Rail Fastenings

1. Introduction to Railway Engineering

Railway = a guided land transport system in which vehicles (rolling stock) run on steel rails. It is the cheapest mode for **bulk goods** and **long-distance mass passenger** movement.

- First train in India: **16 April 1853**, Bombay (Bori Bunder) to Thane, 34 km.
- Indian Railways is one of the largest networks in the world (about 68,000 route-km), run by the Ministry of Railways through the Railway Board and 18 zonal railways.
- A railway system has 4 parts: **(i) track / permanent way, (ii) rolling stock (locomotives, coaches, wagons), (iii) signalling & interlocking, (iv) stations & yards.**

Advantages of railways

- Very economical for heavy and bulk transport; low rolling resistance (steel on steel).
- Safe, reliable, all-weather and energy efficient; less air pollution per tonne-km.
- Can carry large passenger volumes at high speed on one corridor.

Limitations

- High initial cost; fixed routes, so no door-to-door service; needs road feeder links.

2. Permanent Way (Track)

Permanent way is the finished track structure – rails, sleepers, ballast and fastenings – laid on the formation to carry the train load safely. It is called 'permanent' to distinguish it from the temporary track laid during construction.

Components of Permanent Way

- **Rails** – steel girders that guide and carry the wheels.
- **Sleepers** – hold rails at correct gauge and transfer load to ballast.
- **Ballast** – broken-stone bed under sleepers; spreads load, drains water.
- **Fastenings** – fish plates, bolts, spikes, keys, chairs, bearing plates, clips.
- **Formation (sub-grade)** – prepared soil on which ballast rests.

Requirements of an ideal permanent way

- Correct gauge, alignment and level (except cant); uniform and sufficiently **resilient**.
- Adequate strength to bear loads and lateral forces; good drainage; easy renewal.
- Low maintenance cost and provision for expansion and creep control.

Cross-section of permanent way

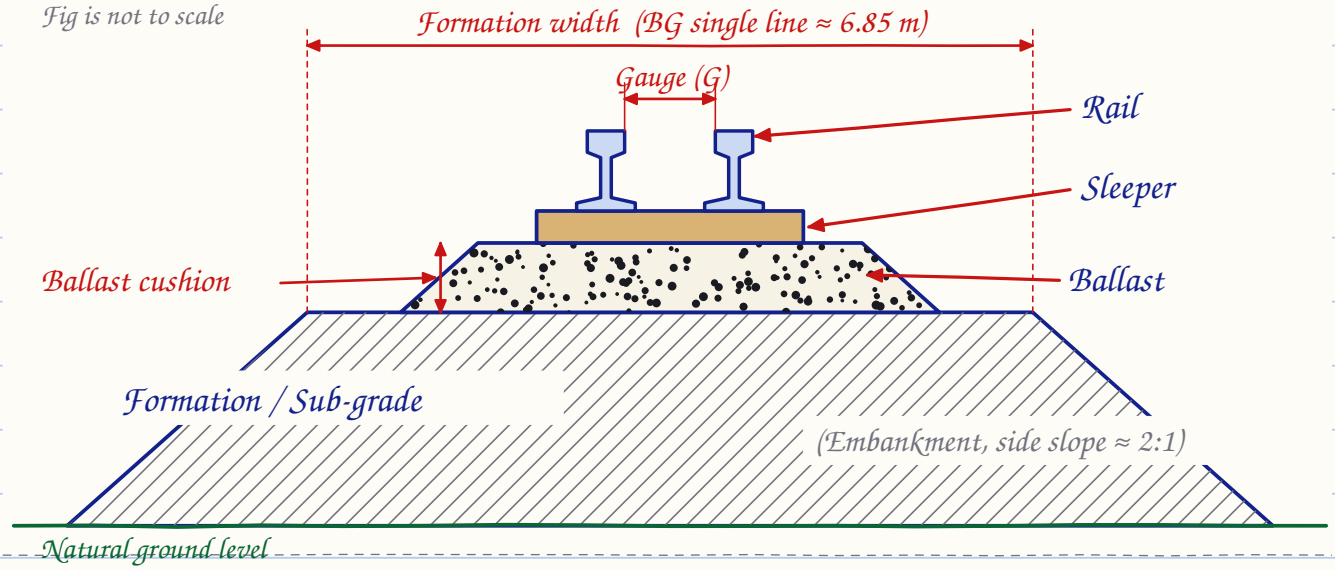


Fig: Cross-section of permanent way (single line, embankment)

- **Formation width** (BG single line embankment ≈ 6.85 m; double line ≈ 11.15 m).
- **Ballast cushion** (below sleeper) 250 – 300 mm; ballast section width ≈ 3.35 m (BG).
- Embankment side slope about 2:1 (H:V); cutting side slope about 1:1 to 1.5:1 – depends on soil.

3. Rails

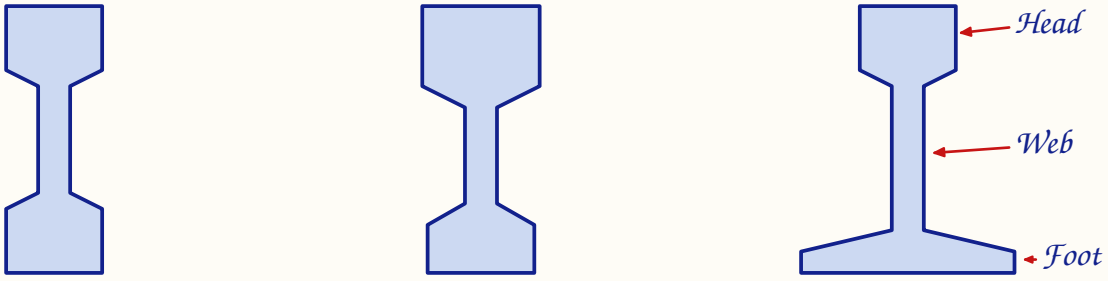
Functions of rails

- Provide a **hard, smooth and continuous surface** for wheels with minimum friction.
- **Guide** the wheels (lateral guidance) and bear the stresses caused by moving loads.
- Transmit the load to the sleepers and ballast over a large area.
- Carry braking and traction forces; act as conductor for track circuits / return current.

Requirements of an ideal rail

- Adequate **vertical stiffness** (depth) and lateral stiffness (head and foot width).
- **Hard head** for wear resistance but tough web/foot against fracture; good fatigue life.
- Weldable, corrosion resistant, low-cost, easy to roll and replace.
- Head wide enough to give wheel contact; foot wide enough for stability.

Rail sections (shape only)



(a) Double headed (b) Bull headed (c) Flat footed

Fig: Types of rail sections

Type	Features	Use
Double headed (DH)	Both heads same, rail can be turned over; needs chairs. Obsolete.	Early railways
Bull headed (BH)	Larger head, smaller foot; cast-iron chairs + keys needed	Britain, sidings
Flat footed (FF) – Vignoles	Wide foot rests directly on sleeper/plate; stiff, cheap, no chairs	Standard in India (~99%)

- Rail sections in India: **90 R (45 kg/m), 52 kg/m, 60 kg/m (UIC 60)** on BG. Heavier rail for higher speed and axle load.
- Standard rail length: **13 m (BG)**, 12 m (MG). Long Welded Rails (LWR) are used for

smooth, low-maintenance track.

- Rail steel: high carbon steel (C = 0.55 – 0.68 %, Mn = 0.7 – 1.0 %); IR uses grade **880** (min UTS 880 N/mm²) and 1080 head-hardened rail.
- Rail defects: wear (vertical/lateral), corrugation, shelling, corrosion at foot, fatigue fracture.

Expansion gap in rails

$$\text{Gap } \Delta = \alpha \cdot L \cdot \Delta T$$

$\alpha = 11.5 \times 10^{-6}$ per °C for steel, L = rail length, ΔT = temperature variation. **Ex:** $L = 13$ m, $\Delta T = 50°C \rightarrow \Delta = 11.5\times10^{-6} \times 13000 \times 50 =$ **7.5 mm**.

4. Sleepers

Functions of sleepers

- Hold the rails to **correct gauge** and keep them in position (alignment & tilt 1 in 20).
- Receive load from rails and **distribute** it over a wider ballast area.
- Act as an elastic medium between rail and ballast; absorb shock and vibration.
- Give lateral stability to the track and keep the rails at proper level (cant on curves).

Requirements of good sleepers

- Initial and maintenance cost low; sufficient bearing area, weight and strength.
- Should allow easy fixing / removal of fastenings and rails; easy to pack with ballast.
- Must resist rot, wear, insects and corrosion; electrically insulating (for track circuits).
- Should be able to be reused; fit for tamping and give good grip with ballast.

Types of sleepers

Type	Merits	Demerits
Wooden (teak, sal, deodar)	Easy handling, resilient, cheap fixing, good insulation, suits all rails	Decay, termite attack, short life (12-20 yr), fire risk
Cast iron (pot, plate, CST-9)	Long life, good scrap value, good gripping	Brittle, heavy, breaks at joints, gauge difficult to hold
Steel (trough, key type)	Light, long life, easy to manufacture, good gauge holding	Corrodes, rail seat loosens, noisy, poor insulation

Type	Merits	Demerits
Concrete (RCC, PSC)	Very long life (40+ yr), heavy, stable, fire-proof, good lateral grip	Heavy, no resilience, needs rubber pad, damages if derailed

- PSC mono-block sleeper (2.75 m, ~270 kg) is the standard on Indian BG main lines; wooden BG sleeper = 2.74 m × 250 mm × 130 mm.

Sleeper density

$$N = M + x$$

N = number of sleepers per rail length, M = rail length in metres, $x = 3$ to 7 (more for heavy/fast traffic). Ex: 13 m rail, $x = 7 \rightarrow N = 20$ sleepers per rail (≈ 1660 per km, spacing ≈ 60 cm).

5. Ballast

Functions of ballast

- Provides a level, hard bed for sleepers; transmits and distributes load to the formation.
- Holds the sleepers in position against lateral and longitudinal movement.
- Gives elasticity and resilience, reducing shock and noise.
- Provides drainage (quick run-off of rain water) and allows packing to adjust track level.
- Checks growth of weeds; helps in maintaining correct super-elevation.

Requirements of good ballast

- Hard, tough, durable, angular with sharp edges (interlocking); no powder/clay.
- Non-porous, should not become slippery when wet; economical and easy to pack (tamping).
- Size ≈ 50 mm (40-65 mm) on main lines; smaller near points & crossings. As per RDSO: Aggregate Impact Value < 20 %, Los Angeles Abrasion < 30 %.

Materials used as ballast

Material	Remarks
Broken stone	Best ballast; used on all main lines (trap, granite, quartzite)
Gravel / river shingle	Rounded, less interlocking; used on branch lines
Cinders / ashes	Cheap and good drainage; corrodes steel, used on sidings

Material	Remarks
Sand	Good drainage, but blows away; used with steel/CI sleepers
Moorum / brick ballast	Poor, low-cost; for lightly loaded tracks, bricks may powder
Selected earth	For temporary track only

- A **sub-ballast / blanket** layer (sand/moorum) is provided on the formation to prevent mud-pumping.

6. Gauges of Track

Gauge = minimum **clear distance between the inner (running) faces** of the two rails of a track.

Standard gauge (Stephenson gauge) = **1435 mm**.

Gauge	Width	Use in India
Broad Gauge (BG)	1676 mm (5'6")	Main lines (≈ 90%+ of network)
Metre Gauge (MG)	1000 mm (3'3⅜")	Branch lines, mostly converted to BG
Narrow Gauge (NG)	762 mm (2'6") / 610 mm (2')	Hill railways, e.g. Darjeeling, Nilgiri, Kalka–Shimla
Standard Gauge	1435 mm	Metro / world standard; used in some Indian metros

Factors in selecting a gauge

- Volume of traffic and speed required; **cost** of construction (wider gauge costs more).
- Physical features – hilly area needs sharper curves, so narrow gauge; future development.
- Uniformity with adjoining systems (Unigauge policy).

Problems of multiple gauges (break of gauge)

- Transhipment of goods and passengers, delay, damage and extra labour cost.
- Extra rolling stock and duplicate workshops/stations needed at junctions; hence the **Unigauge project**.

7. Creep of Rails

Definition

Creep is the gradual longitudinal movement of rails with respect to the sleepers, normally in the direction of dominant traffic. Rails creep forward, not sleepers.

Indications of creep

- Closing of expansion gaps at one end and opening at the other; joints out of square.
- Marks (scratches) on sleepers/ chairs where rail moved; rails shifting past the creep indicator (white paint / stake).
- Kinks in alignment, sleepers out of square, rails shearing fish bolts.

Causes of creep

- Wave action of rail under moving load; drag of wheels; percussion at rail ends.
- Starting, accelerating and braking of trains; temperature changes.
- Loose fastenings, poor ballast, loose joints, sharp curves, steep gradients (more on down gradient).
- Creep is more on heavy-traffic lines, curves, approach to stations, bridges, and with light rail sections.

Theories related to creep

Theory	Explanation
Wave theory	Wheel load makes a depression in the rail like a wave; the wave moves ahead with the wheel and pushes the rail forward, as the rail is elastic between sleepers.
Drag theory	Tractive force from driving wheels (and braking) drags the rail forward by friction between wheel and rail.
Percussion theory	Wheel strikes the end of the next rail (at joint) and gives a forward blow; the energy pushes the rail ahead, so creep is greater at joints.
Temperature (other) causes	Expansion in warm weather and contraction in winter, plus vibration, cause rails to shift.

Effects of creep

- Sleepers get out of square → gauge and alignment disturbed; points and crossings are disturbed.
- Expansion gaps lost → buckling (sun kinks) in summer; joints open up and fish bolts shear.
- Ballast disturbed; increased cost of maintenance; derailment risk.

Prevention / correction of creep

- Pulling back the rails to original position; use anchors / anti-creep devices and good fastenings.
- Keep ballast compact (full), use heavier rail and well-packed sleepers; tighten fish bolts; use long welded rails.
- Use steel / concrete sleepers with elastic clips (Pandrol, ERC) which grip the rail foot.
- Creep is measured with creep indicators; action is needed if creep exceeds about 150 mm (BG).

8. Coning of Wheels and Tilting of Rails

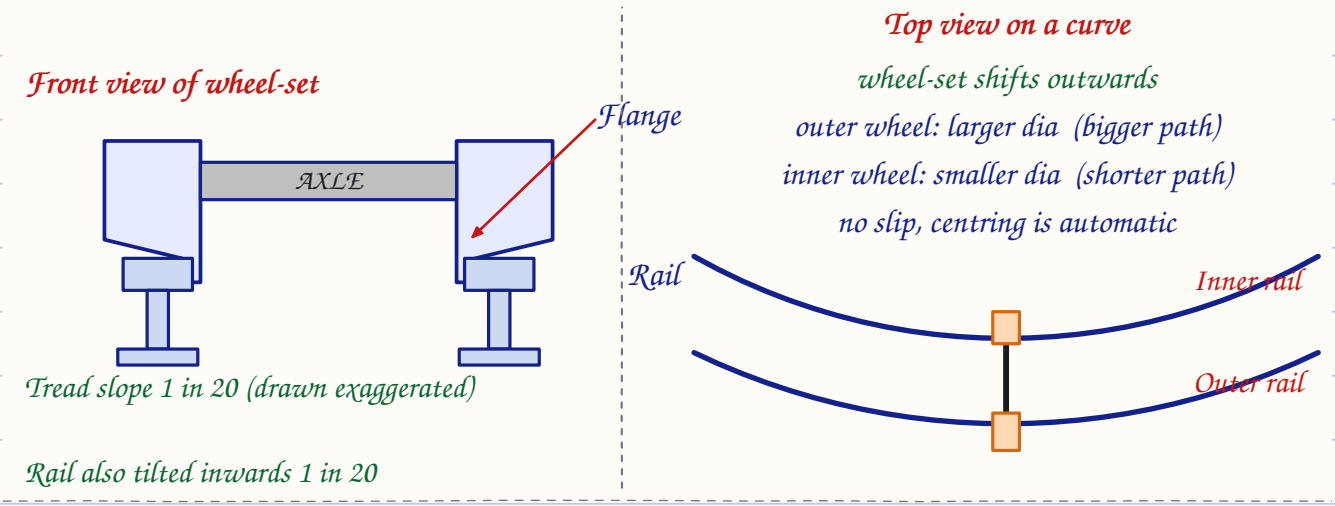


Fig: Coning of wheels and rail tilt

Coning of wheels

Treads of wheels are made conical with a slope of 1 in 20 (not flat) – larger diameter at the flange side. The rails are also tilted inwards 1 in 20 so that the rail head fits the wheel tread.

Reasons / advantages of coning

- Wheel treads wear **uniformly**, which prevents formation of a groove in the rail head.
- Provides **lateral movement** – when the wheel-set shifts to one side, the wheel on that side rolls on a larger diameter and the other on a smaller one, so it **recentres automatically** (self-correcting).
- On curves, the outer wheel gets a larger diameter and travels the longer outer path **without slip**; flange wear and friction are reduced.
- Gives smooth riding, reduces wear of rails and flanges, and checks sway.

Coning on a curve (derivation)

For no slip: $(r + x/20) / (r - x/20) = (R + G/2) / (R - G/2)$, so the lateral shift is:

$$x = 10 \cdot G \cdot r / R$$

Ex: $G = 1.7$ m, wheel radius $r = 0.45$ m, $R = 500$ m $\rightarrow x = 10 \times 1.7 \times 0.45 / 500 = 0.0153$ m $\approx$ **15 mm** (within the flange clearance, so the curve is negotiated without slip).

- If the required shift exceeds flange clearance, flange touches the rail and gauge widening / speed restriction is needed.
- Reason for **tilting rails**: if rails are laid vertical and wheels coned, contact would be at the gauge corner – heavy wear. Tilt 1 in 20 brings contact to the head centre, reduces eccentric load.

9. Adzing of Sleepers

Definition

Adzing is the cutting (levelling) of the rail seat on a wooden sleeper with an **adze** so that the seat slopes **1 in 20 inwards**, giving the required **tilt to the rail**.

- Done only on **wooden sleepers** (the tilt is built in the shape of steel/CI/concrete sleepers and plates).
- Seat cut is just enough to take the rail foot / bearing plate; seats of both rails are cut to slope towards each other.
- Alternative: use **canted bearing plates** (cast / MS plates with 1 in 20 slope) – then adzing is not necessary.

- Disadvantage: adzing removes the hard top layer of the sleeper, so it weakens the sleeper, makes it rot early and shortens its life; hence timber is treated (creosote) after adzing.

10. Rail Fastenings

Fastenings hold the rails to each other (rail joints) and rails to sleepers. They should be strong, simple, elastic and easy to remove.

(a) Fastenings at rail joints

- Fish plates – two per joint, bolted through web to join rail ends; 4-hole (and 6-hole for heavy rail), slope of 1 in 4 matches fish-plate seat.
- Fish bolts (with nuts and spring washers) – 4 per joint; keep the plates tight and allow expansion.
- Joints are suspended, supported or bridge; welding (SWR, LWR/CWR) removes joints.

(b) Fastenings between rail and sleepers

Fastening	Purpose / Note
Dog spike / round spike	Drives into wooden sleeper; holds rail foot or plate (rail-sleeper); screw spike gives better grip
Fang bolt & hook bolt	Used with steel / trough sleepers to fix rail
Keys (wooden, steel)	Wedge to hold rail in the chair (BH rails), with chairs
Chairs (cast iron)	Support the bull-headed rail on sleepers
Bearing plates (MS / CI)	Spread load over a larger sleeper area, give 1 in 20 cant, avoid adzing
Elastic fastening (Pandrol clip, ERC, rubber pad)	Grips the rail foot with spring force, resists creep, resilient; used on PSC sleepers
Anchors / anti-creep	Fix rail to sleepers to resist creep

11. Rail Joints and Welded Rails

A **rail joint** is the place where two rail ends are connected with fish plates. It is the **weakest point** of the track – it causes noise, impact (hammering), wear, creep and needs frequent maintenance.

Type of joint	Description
Supported joint	Rail ends rest directly on one sleeper (joint sleeper); sleeper gets heavy blows.
Suspended joint	Rail ends lie between two sleepers (placed close, 300-450 mm apart); most common on IR; gives elasticity.
Bridge joint	Rail ends joined by a metal bridge plate fixed on two or more sleepers; sturdy but costly.
Staggered / square joint	Staggered: joints of two rails not opposite; square: joints exactly opposite each other (used in India).

Welded rails

- **SWR** (short welded rail, 3 rails = 39 m), **LWR** (long welded rail, > 250 m) and **CWR** (continuous welded rail – a whole section).
- Advantages: no joints → smooth ride, less noise, less wear, **lower maintenance cost**, longer life of rail & rolling stock.
- In LWR the central portion is **fully anchored** and stays constant length; breathing lengths (end portions) move; temperature stresses are taken by the rail itself.
- Welding methods: **flash butt**, **thermit (aluminothermic)** and gas pressure welding; rail temperature is fixed (destressing).
- Care needed: buckling risk in summer; creep control by proper fastenings and full ballast.

12. Worked Examples

Example 1 – Expansion gap

Rail length $L = 13$ m; temperature range $\Delta T = 40°C$; $\alpha = 11.5 \times 10^{-6}$ /°C.

Gap $\Delta = \alpha L \Delta T = 11.5 \times 10^{-6} \times 13000 \times 40 =$ **5.98 mm ≈ 6 mm**.

Example 2 – Number of sleepers

Length of track = 1 km, rail length 13 m, $x = 7 \rightarrow N = 13 + 7 = 20$ per rail.

Rails per km = 1000 / 13 = 76.9 ≈ 77. Sleepers = 77 × 20 = **1540** (about 1540–1660 per km depending on density).

Example 3 – Lateral shift on a curve

$G = 1.75$ m, $r = 0.46$ m, $R = 700$ m. $x = 10 G r / R = 10 \times 1.75 \times 0.46 / 700 = 0.0115$ m = **11.5 mm**.

Example 4 – Ballast volume (approx.)

Ballast width at top 2.25 m, bottom 3.35 m, depth 0.25 m, length 1 km.

Area = (2.25 + 3.35)/2 × 0.25 = 0.70 m²; Volume = 0.70 × 1000 = **700 m^3 per km** (sleeper volume is deducted for exact value).

13. Quick Comparison Tables

Point	Wooden sleeper	Concrete sleeper
Life	12-20 years	40+ years
Weight	Light (~ 80 kg)	Heavy (~ 270 kg)
Elasticity	Good	Poor – needs pad
Fastening	Spikes / screws	Elastic clips (ERC / Pandrol)
Adzing	Required (or canted plate)	Not required (slope built in)
Cost / maintenance	Low initial, high upkeep	High initial, low upkeep

Theory of creep	Main idea	Key word
Wave	Rail moves ahead as wave pushes	Wave / depression
Drag	Tractive force of wheel drags rail	Friction

Theory of creep	Main idea	Key word
Percussion	Wheel hits rail end at joint	Impact

Remember (one-line facts)

- Permanent way = rails + sleepers + ballast + fastenings.
- Gauge is measured between inner faces of the rails.
- Creep occurs in the direction of traffic; it is more on down gradients and at braking zones.
- Coning: wheel tread 1 in 20; rail tilt: 1 in 20 by adzing or canted bearing plates.
- Fish plate slope 1 in 4; max permitted gap about 20 mm (13 m rail).

Important Questions – Unit I

Short answer questions (2 marks)

- Define permanent way and list its components.
- What is gauge? Name the gauges used in India.
- What is coning of wheels? What is the slope?
- Define creep. What is adzing?
- Write the sleeper density formula.

Long answer questions (10 marks)

- Draw a neat cross-section of a permanent way and explain the functions of its components.
- Explain the requirements and types of sleepers with merits and demerits.
- What are the functions of ballast? Explain the requirements and materials of ballast.
- Explain creep and the theories related to creep; give causes, effects, and preventive measures.
- Explain coning of wheels and adzing of sleepers; Discuss various rail fastenings.

Quick revision: Gauge BG 1676, MG 1000, NG 762/610, Std 1435 mm | Cone & tilt = 1 in 20 | $N = M + x$ ($x = 3–7$) | Ballast cushion 250–300 mm | Rail length 13 m.